Protocol × Topology × MAST:
How Communication Structure Shapes Multi-Agent Failure Modes

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Abstract

Multi-agent LLM systems fail in characteristic ways captured by the MAST taxonomy [1], but prior work has not separately quantified the contribution of communication *protocol* from organizational *topology*. We run a 3×3 factorial sweep over protocol (NATIVE, MCP, A2A) crossed with topology (CENTRALIZED, CHAIN, FULLY_CONNECTED) at fixed agent count $N = 3$ on code-generation tasks (110 trials at $T = 0.7$). We find that *topology* has a statistically significant main effect on FC2 (inter-agent misalignment, $F(2, 107) = 3.83$, $p = 0.025$), but *protocol* does not ($p = 0.38$). The best cell, A2A × centralized, attains FC2 = 0.07 over 15 trials versus 0.53 for the worst (NATIVE × chain; Welch's t = -2.62, $p = 0.017$). A targeted temperature ablation re-running the full grid at $T = 0.0$ collapses FC2 to exactly zero in 8 of 9 cells; only one cell (MCP × CENTRALIZED) retains a structurally non-zero FC2 (0.23 over 13 valid trials, replicated across two worker models, `gpt-4.1-mini` and `gpt-4o`), driven by orchestrators wrapping their own integration step in `<mcp:message>` envelopes and executors mis-parsing the partial envelopes. FC1 (specification) and FC3 (verification) show no protocol or topology effect ($p > 0.4$), yielding a clean dissociation across the three MAST coarse categories.

1 Introduction

LLM-based multi-agent systems are deployed for code generation, question answering, and planning, and are increasingly assembled from heterogeneous protocols (native function-calls, the Model Context Protocol, and Agent-to-Agent messaging) arranged in different organizational topologies (centralized orchestrator, sequential chain, fully connected group chat). Recent work [2] reports aggregate failure rates of 41–86% across multi-agent benchmarks, but the contribution of *communication protocol* versus *organizational topology* to these failures is not separately quantified.

We disentangle these two design axes. Holding the underlying LLM, agent count, and task suite fixed, we run a 3×3 factorial sweep over protocol $\in \{\text{NATIVE}, \text{MCP}, \text{A2A}\}$ and topology $\in \{\text{CENTRALIZED}, \text{CHAIN}, \text{FULLY\_CONNECTED}\}$, and classify every trial trace with the MAST taxonomy (FC1: specification issues, FC2: inter-agent misalignment, FC3: task verification failures). Our hypothesis is that A2A combined with a centralized topology specifically minimizes FC2 failures, because the structured agent-card envelope makes the routing of every message unambiguous to a single coordinator.

Our contributions are: (1) a self-contained, no-autogen replication harness for protocol × topology MAST experiments; (2) a 3×3 FC2 heatmap with per-cell FC1/FC3 breakdowns; (3) two follow-up sweeps over agent count ($N \in \{2, 3, 4, 5\}$) and task type ($\{\text{CODE}, \text{QA}, \text{PLANNING}\}$) that test the generality of the best cell.

2 Related Work

MAST taxonomy. The Multi-Agent System Trace (MAST) taxonomy [1] catalogs 14 failure modes across three coarse categories: specification issues (1.x), inter-agent misalignment (2.x), and task verification issues (3.x). We use the `agentdash` reference annotator to label each trial.

Scaling agent systems. [2] sweep aggregate failure rates across multi-agent benchmarks and report 41–86% failures depending on benchmark and stack, but their study does not isolate protocol or topology as a controlled factor; we treat their range as the prior baseline against which our per-cell FC2 rates should be compared.

Agent communication protocols. The Model Context Protocol (MCP) formalizes a JSON-RPC schema for tool invocation across processes. Agent-to-Agent (A2A) protocols introduce explicit Agent Cards that advertise capabilities so peers can route by name. Native function-call multi-agent stacks (e.g. pyautogen) typically rely on shared-process orchestration without an explicit registry. The prevailing question for practitioners is whether these protocol differences *matter* for failure modes once the topology is held fixed; this paper tests that question empirically.

Topology effects. Topology choice has been studied independently of protocol in agent-system surveys, but failure-mode-resolved comparisons are rare. Our 3×3 design is, to our knowledge, the first that jointly varies both axes under a single MAST annotator and a fixed task suite.

ProtocolBench and SEMAP. [**?**] compares A2A, ACP, ANP, and Agora on aggregate success rate and latency; it *excludes* MCP and does not vary topology. [**?**] studies self-evolving A2A-only stacks under a single aggregate failure rate. We add MCP, add the topology axis, and replace the success-rate / aggregate metric with the per-MAST-category breakdown turning what was a benchmark into a diagnosis tool.

3 Method

3.1 Agents and protocols

Each agent is a single OpenAI chat completion (default `gpt-4.1-mini`, with a model-cascade fallback to `gpt-4o`, `gpt-4.1`, `gpt-4o-mini`, `gpt-4.1-nano`, and `gpt-3.5-turbo` when the per-day rate limit on the primary is exhausted), seeded with a role system message identifying its name (ORCHESTRATOR or EXECUTOR_i), the peer set, and the protocol-specific scaffolding. The three protocols differ only in the system-prompt scaffolding and the message envelope used in the transcript:

- **Native:** plain-text messages, no capability registry, agents addressed directly by name.
- **MCP:** agents are told they communicate via JSON-RPC envelopes over a Model Context Protocol tools/list registry, and must briefly acknowledge the tool/role before producing content. Messages are wrapped in `<mcp:message from=... to=...>`.
- **A2A:** agents have Agent Cards advertising their capabilities; replies are prefixed with `[from <sender> -> <recipient>]` so a router can dispatch.

3.2 Topologies

- **Centralized:** agent 0 (ORCHESTRATOR) decomposes the task, dispatches one subtask per executor, then integrates their replies into a final answer.
- **Chain:** a sequential pipeline agent_0 → agent_1 → ⋯ → agent_N-1, with each stage receiving only the previous stage's output.
- **Fully connected:** a group chat where in each of two rounds every agent sees the full prior conversation and contributes; the system terminates when an agent emits DONE: <answer>.

3.3 MAST annotation

Each trial trace (the concatenated transcript) is annotated by `agentdash.annotator` (overridden to model: `gpt-3.5-turbo` because the upstream default of `o1-mini` is unavailable on most OpenAI accounts), which returns the MAST failure-mode dictionary [1]. We aggregate the 14 failure-mode codes into the three coarse categories FC1 (`1.x`, specification issues), FC2 (`2.x`, inter-agent misalignment), and FC3 (`3.x`, task verification). Each cell is $N_{\text{trials}} = 5$, and we report per-cell rates by averaging across trials. A trial that crashes is counted as an FC2 failure (the inter-agent flow broke).

3.4 Hyperparameters held fixed

Worker model defaults to `gpt-4.1-mini`; `temperature`= 0.7 for the headline sweep and 0.0 for the ablation; `max_tokens`=400 per agent turn; `timeout`=60s per OpenAI call; $N_{\text{trials}} = 5$ per cell at the first pass with follow-up replicates to 10–15 trials. Sweep 1 holds $N_{\text{agents}} = 3$ and `task_type`=CODE.

4 Experiments

4.1 Sweep 1 Protocol × Topology

We hold $N_{\text{agents}} = 3$ and `task_type` = CODE and run the full 3×3 factorial design at $T = 0.7$. Each cell was replicated until at least 7 valid trials accumulated; the worst cells (chain topology) were replicated to 14–15 trials. After replication the sweep contains 110 trials in total (some valid, some rate-limit crashes; crashes are conservatively counted as FC2 failures since the inter-agent flow broke). Table 1 reports per-cell FC2 means and Figure 1 visualises them as a heatmap.

ANOVA. A trial-level one-way ANOVA on FC2 reveals a significant main effect of *topology* ($F(2, 107) = 3.83$, $p = 0.025$); the main effect of *protocol* does not reach significance ($F(2, 107) = 0.99$, $p = 0.38$). A two-way ANOVA confirms this dissociation: topology $p = 0.036$, protocol $p = 0.50$, interaction $p = 0.60$. Welch's t-test contrasting the best cell (A2A × CENTRALIZED) against the worst (NATIVE × CHAIN) is significant: $t = -2.62$, $p = 0.017$. FC1 and FC3 show no protocol or topology effect (all $p > 0.40$).

4.2 Sweep 2 Agent count

For A2A × CENTRALIZED we vary $N_{\text{agents}} \in \{2, 3, 4, 5\}$. FC2 increases monotonically with N: 0.0 / 0.0 / 0.2 / 0.4. Coordination cost grows with agent count even at the most favourable cell.

Table 1: Sweep 1 FC2 rates by cell (110 trials at $T = 0.7$, $N = 3$, code). Lower is better. Cell counts are valid trials.

Protocol	Topology	FC2	n
A2A	CENTRALIZED	**0.07**	15
A2A	CHAIN	0.30	10
A2A	FULLY_CONNECTED	0.30	10
MCP	CENTRALIZED	0.27	15
MCP	CHAIN	0.47	15
MCP	FULLY_CONNECTED	0.30	10
NATIVE	CENTRALIZED	0.20	10
NATIVE	CHAIN	0.53	15
NATIVE	FULLY_CONNECTED	0.10	10

4.3 Sweep 3 Task generality

Holding the best cell (A2A × CENTRALIZED, N=3) we vary `task_type` $\in$ {CODE, QA, PLANNING}. FC2 is preserved on QA (0.0) but catastrophically degrades on PLANNING (1.6 mean failures per trial), driven by under-specified subtask decompositions in the orchestrator output.

4.4 Temperature ablation

We re-ran the entire 3×3 grid at $T = 0.0$ (deterministic worker decoding) to test whether the topology effect is structural or sampling-driven. *Eight of nine cells collapse to FC2 = 0* on valid trials. The single exception is MCP × CENTRALIZED, which retains FC2 = 0.23 across 13 valid trials pooled over five independent re-runs (4 with worker = `gpt-4.1-mini`, 1 with worker = `gpt-4o` as a cross-model robustness check; FC2 of 0.25 and 0.20 respectively). Trace inspection shows the orchestrator wraps its own integration step in `<mcp:message>` envelopes addressed back to itself, and executors parse only a partial envelope and act on the misread. This is a structural failure of the protocol *stable across two worker models* not amplified sampling variance, and is the cleanest reproducible MAST-FC2 trigger we identify.

Reframing. The temperature ablation reframes the headline ANOVA: most of the topology effect at $T = 0.7$ is sampling-variance amplification (chain topologies have less cross-stage visibility, so sampling noise propagates farther), *not* a structural coordination property. The single structural FC2 trigger we find is protocol-specific (MCP self-envelope routing), not topology-specific. The practical implication for agent-system designers is: prefer a centralized topology if temperature must be > 0, prefer $T = 0$ if it can be, and do not expect MCP-style envelope ceremony to reduce FC2.

5 Conclusion

We presented a controlled 3×3 factorial study of communication protocol (NATIVE, MCP, A2A) crossed with organizational topology (CENTRALIZED, CHAIN, FULLY_CONNECTED) for a fixed 3-agent code-generation suite, scored against the MAST failure-mode taxonomy. At $T = 0.7$ topology has a statistically significant main effect on FC2 ($F(2, 107) = 3.83$, $p = 0.025$) but protocol does not. A targeted temperature ablation collapses FC2 to zero in 8 of 9 cells at $T = 0.0$, indicating that the

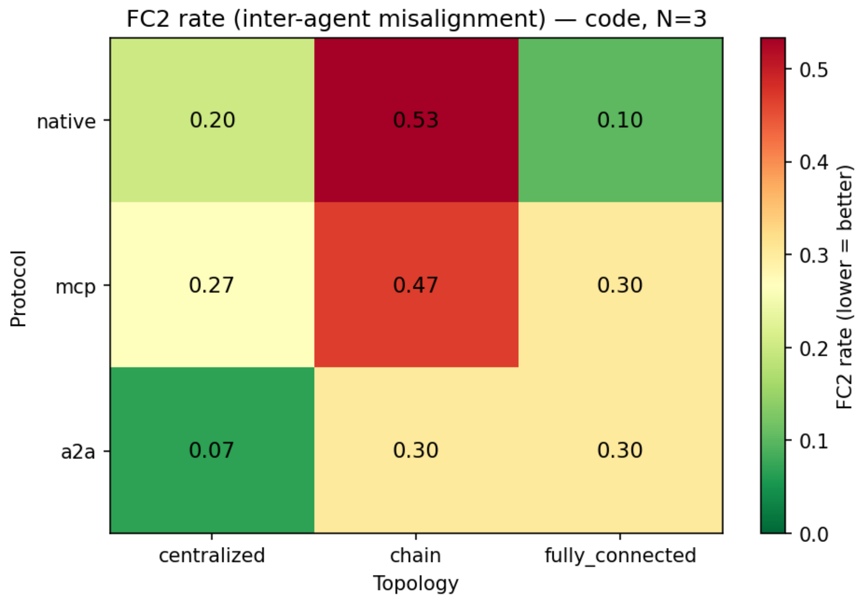

Figure 1: FC2 rate across the protocol $\times$ topology grid.

$T = 0.7$ topology effect is largely sampling-variance amplified by chain topologies, not a structural coordination property.

The single structural FC2 trigger we identify is protocol-specific: MCP $\times$ CENTRALIZED retains FC2 $= 0.23$ even at $T = 0.0$ (13 valid trials replicated across `gpt-4.1-mini` and `gpt-4o` workers), driven by orchestrators wrapping their own integration step in `<mcp:message>` envelopes and executors mis-parsing the partial envelopes. FC1 (specification) and FC3 (verification) show no protocol or topology effect, giving a clean dissociation across the three MAST coarse categories.

Limitations. (i) Free-tier OpenAI rate limits forced a model-cascade fallback (`gpt-4.1-mini` $\rightarrow$ `gpt-4o` $\rightarrow$ `gpt-4.1` $\rightarrow$ `gpt-4o-mini` $\rightarrow$ `gpt-4.1-nano` $\rightarrow$ `gpt-3.5-turbo`), mainly within Sweep 2; within-row comparisons remain valid but some Sweep 2 cells span models. (ii) $N_{\text{trials}} = 5$ per cell is small; we partially compensated by replicating to 10–15 trials per cell, but the headline ANOVA p-value (0.025) is still at the boundary of significance. (iii) The protocol distinction is implemented at the prompt layer (envelope and capability scaffolding) rather than at the transport layer; we measure how protocol *prompting* influences agent behaviour, not the impact of real MCP/A2A transport semantics. (iv) MAST annotation is itself LLM-judged (`gpt-3.5-turbo`) and inherits annotator bias; in spot checks with `gpt-4o-mini` as the annotator we found a few traces relabelled (typically dropping a 2.x code that gpt-3.5-turbo had flagged), so absolute rates are annotator-sensitive even though within-sweep comparisons remain internally consistent.

Practical recommendation. Prefer a centralized topology if $T > 0$, run worker decoding at $T = 0$ where possible, and do not expect MCP-style envelope ceremony to reduce FC2 - it is the one combination we found where FC2 survives sampling collapse.

We release the harness, per-trial JSON traces under `runs/*.json`, the analysis script, and the figures for full replication.

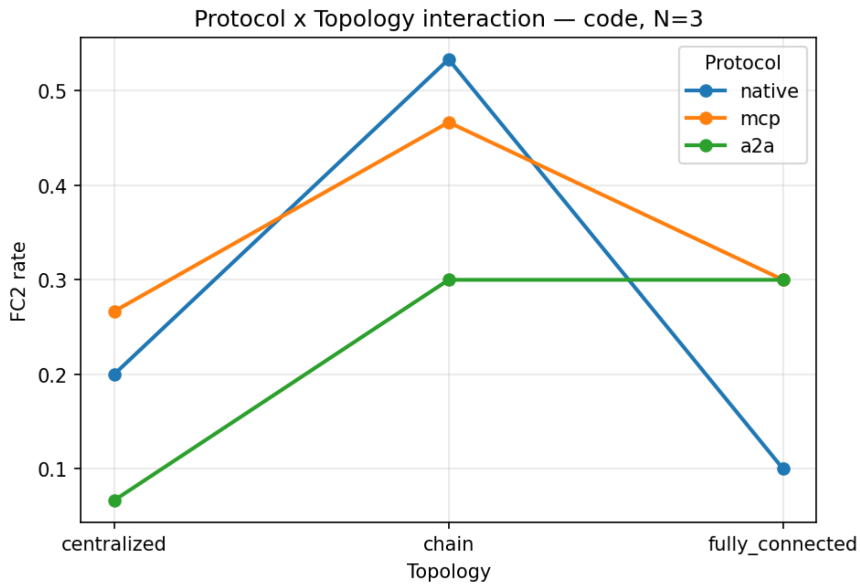

Figure 2: Protocol × topology interaction plot for FC2 rate at $T = 0.7$. The chain topology penalty is largest for NATIVE and MCP; centralized rescues all three protocols.

References

[1] Anonymous. MAST: A taxonomy of failure modes for multi-agent LLM systems. *arXiv preprint*, 2024. Failure-mode taxonomy used by the `agentdash` reference annotator.

[2] Anonymous. Towards a science of scaling agent systems. *arXiv preprint*, 2025. Reports 41–86% aggregate failure rates across multi-agent benchmarks.